

Praticial-1

```
D:\>mkdir microservices-demo
D:\>cd microservices-demo
D:\microservices-demo>mkdir backend
D:\microservices-demo>cd backend
D:\microservices-demo\backend>npm init -y
Wrote to D:\microservices-demo\backend\package.json:

{
  "name": "backend",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}

D:\microservices-demo\backend>npm install express mongoose
added 82 packages, and audited 83 packages in 4s

23 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities

D:\microservices-demo\backend>
```

```
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}

D:\microservices-demo\backend>npm install express mongoose
added 82 packages, and audited 83 packages in 4s

23 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities

D:\microservices-demo\backend>notepad server.js
D:\microservices-demo\backend>notepad Dockerfile
D:\microservices-demo\backend>ren Dockerfile.txt Dockerfile
D:\microservices-demo\backend>cd..
D:\microservices-demo>notepad docker-compose.yml
D:\microservices-demo>docker-compose up --build
time="2026-04-19T12:46:53+05:30" level=warning msg="D:\\microservices-demo\\docker-compose.yml: the attribute 'version' is obsolete,
it will be ignored, please remove it to avoid potential confusion"
#1 [internal] load local bake definitions
#1 reading from stdin 541B done
#1 DONE 0.0s

#2 [internal] load build definition from Dockerfile
#2 transferring dockerfile: 150B done
```

```

D:\microservices-demo\backend>cd..

D:\microservices-demo>notepad docker-compose.yml

D:\microservices-demo>docker-compose up --build
time="2026-04-19T12:46:53+05:30" level=warning msg="D:\\microservices-demo\\docker-compose.yml: the attribute 'version' is obsolete,
it will be ignored, please remove it to avoid potential confusion"
#1 [internal] load local bake definitions
#1 reading from stdin 541B done
#1 DONE 0.0s

#2 [internal] load build definition from Dockerfile
#2 transferring dockerfile: 150B done
#2 DONE 0.0s

#3 [internal] load metadata for docker.io/library/node:18
#3 ...

#4 [auth] library/node:pull token for registry-1.docker.io
#4 DONE 0.0s

#3 [internal] load metadata for docker.io/library/node:18
#3 DONE 4.4s

#5 [internal] load .dockerignore
#5 transferring context: 2B done
#5 DONE 0.1s

#6 [1/5] FROM docker.io/library/node:18@sha256:c6ae79e38498325db67193d391e6ec1d224d96c693a8a4d943498556716d3783
#6 resolve docker.io/library/node:18@sha256:c6ae79e38498325db67193d391e6ec1d224d96c693a8a4d943498556716d3783 0.0s done
#6 sha256:461077a72f77fe40d34a37d6a1958c4d16772d0dd77f572ec50a1fdc41a3754d 0B / 446B 0.2s
#6 sha256:3697be50c98b9d071df4637e1d3491d00e7b9f3a732768c876d82309b3c5a145 0B / 1.25MB 0.2s
#6 sha256:c6b30c3f16966552af10ac00521f60355b1fcfd46ac1c20b1038587e28583ce7 0B / 45.68MB 0.2s
#6 sha256:cda7f44f2bdcc4bb7514474024b3f3705de00ddb6355a33be5ac7808e5b7125 0B / 3.32kB 0.2s
#6 ...

#7 [internal] load build context
#7 transferring context: 11.93MB 1.0s done
#7 DONE 1.1s

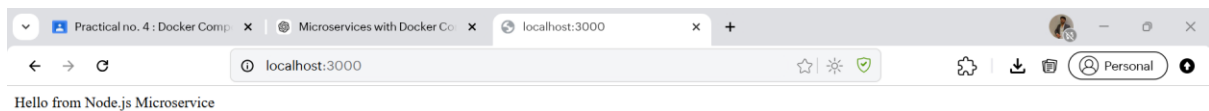
#12 exporting to image
#12 exporting layers
#12 exporting layers 2.5s done
#12 exporting manifest sha256:8c91c6fcfcl90854300632e105f86141287973eee4e4ed4730d9bffc6353c3d4 0.1s done
#12 exporting config sha256:15d91f8e9416227af1e3aa2147107016391655c0aac53e693796347c446ebb60 0.1s done
#12 exporting attestation manifest sha256:40f784b750c9c8065f2ee135300cdd5b603acfc07e9bb918896999b6e63ed7d 0.3s done
#12 exporting attestation manifest sha256:40f784b750c9c8065f2ee135300cdd5b603acfc07e9bb918896999b6e63ed7d 0.3s done
#12 exporting manifest list sha256:el01fde8e5ec178552cc3b8d860695f2df20f35a5e0770449d2cf8212cb4a50
#12 exporting manifest list sha256:el01fde8e5ec178552cc3b8d860695f2df20f35a5e0770449d2cf8212cb4a50 0.1s done
#12 naming to docker.io/library/microservices-demo-backend:latest
#12 naming to docker.io/library/microservices-demo-backend:latest 0.0s done
#12 unpacking to docker.io/library/microservices-demo-backend:latest
#12 unpacking to docker.io/library/microservices-demo-backend:latest 1.1s done
#12 DONE 4.5s

#13 resolving provenance for metadata file
#13 DONE 0.0s
[+] up 5/5
  ✓ Image microservices-demo-backend Built 83.6s
  ✓ Network microservices-demo_default Created 0.1s
  ✓ Volume microservices-demo_mongo_data Created 0.0s
  ✓ Container mongo_db Created 0.4s
  ✓ Container node_app Created 0.3s
Attaching to mongo_db, node_app
node_app | Server running on port 3000
mongo_db | {"t":{"$date":"2026-04-19T07:18:20.621+00:00"},"s":"I", "c":"-", "id":8991200, "ctx":"main","msg":"Shuffling init
ializers","attr":{"seed":527510628}}
mongo_db | {"t":{"$date":"2026-04-19T07:18:20.652+00:00"},"s":"I", "c":"CONTROL", "id":97374, "ctx":"main","msg":"Automatically
disabling TLS 1.0 and TLS 1.1, to force-enable TLS 1.1 specify --sslDisabledProtocols 'TLS1_0'; to force-enable TLS 1.0 specify --ssl
DisabledProtocols 'none'"}
mongo_db | {"t":{"$date":"2026-04-19T07:18:20.654+00:00"},"s":"I", "c":"NETWORK", "id":4915701, "ctx":"main","msg":"Initialized wi
re specification","attr":{"spec":{"incomingExternalClient":{"minWireVersion":0,"maxWireVersion":27},"incomingInternalClient":{"minWir
eVersion":0,"maxWireVersion":27},"outgoing":{"minWireVersion":27,"maxWireVersion":27},"isInternalClient":true}}}
mongo_db | {"t":{"$date":"2026-04-19T07:18:20.661+00:00"},"s":"I", "c":"CONTROL", "id":5945603, "ctx":"main","msg":"Multi threadin
g initialized"}
mongo_db | {"t":{"$date":"2026-04-19T07:18:20.662+00:00"},"s":"I", "c":"CONTROL", "id":4615611, "ctx":"initandlisten","msg":"Mongo
DB starting","attr":{"pid":1,"port":27017,"dbPath":"/data/db","architecture":"64-bit","host":"10eabd1fd604"}}
mongo_db | {"t":{"$date":"2026-04-19T07:18:20.662+00:00"},"s":"I", "c":"CONTROL", "id":23403, "ctx":"initandlisten","msg":"Build
Info","attr":{"buildInfo":{"version":"8.2.5","gitVersion":"a471a13094434666c48a1f75451f2efa49f8f5df","opensslVersion":"OpenSSL 3.0.1
3 30 Jan 2024","modules":[],"allocator":"tcmalloc-google","environment":{"distmod":"ubuntu2404","distarch":"x86_64","target_arch":"x8

```

```
WSL2" "type":"Linux"},"env":{"container":{"runtime":"docker"}}}}}
mongo_db | {"t":{"$date":"2026-04-19T07:18:21.621+00:00"},"s":"I", "c":"ACCESS", "id":10483900,"ctx":"conn1","msg":"Connection not authenticating","attr":{"client":"172.20.0.3:41830","doc":{"driver":{"name":"nodejs|Mongoose","version":"7.1.1|9.4.1"},"platform":"Node.js v18.20.8, LE","os":{"name":"linux","architecture":"x64","version":"6.6.87.2-microsoft-standard-WSL2"},"type":"Linux"},"env":{"container":{"runtime":"docker"}}}}}
mongo_db | {"t":{"$date":"2026-04-19T07:18:21.628+00:00"},"s":"I", "c":"NETWORK", "id":22943, "ctx":"listener","msg":"Connection accepted","attr":{"remote":"172.20.0.3:41840","isLoadBalanced":false,"uuiid":{"$uuiid":{"$uuiid":"57e0693e-200f-40cb-82b5-47d6fc029a1a"}}},"connectionId":2,"connectionCount":2}}
mongo_db | {"t":{"$date":"2026-04-19T07:18:21.629+00:00"},"s":"I", "c":"NETWORK", "id":51800, "ctx":"conn2","msg":"client metadata","attr":{"remote":"172.20.0.3:41840","client":"conn2","negotiatedCompressors":[],"doc":{"driver":{"name":"nodejs|Mongoose","version":"7.1.1|9.4.1"},"platform":"Node.js v18.20.8, LE","os":{"name":"linux","architecture":"x64","version":"6.6.87.2-microsoft-standard-WSL2"},"type":"Linux"},"env":{"container":{"runtime":"docker"}}}}}
mongo_db | {"t":{"$date":"2026-04-19T07:18:21.629+00:00"},"s":"I", "c":"ACCESS", "id":10483900,"ctx":"conn2","msg":"Connection not authenticating","attr":{"client":"172.20.0.3:41840","doc":{"driver":{"name":"nodejs|Mongoose","version":"7.1.1|9.4.1"},"platform":"Node.js v18.20.8, LE","os":{"name":"linux","architecture":"x64","version":"6.6.87.2-microsoft-standard-WSL2"},"type":"Linux"},"env":{"container":{"runtime":"docker"}}}}}
mongo_db | {"t":{"$date":"2026-04-19T07:18:32.131+00:00"},"s":"I", "c":"NETWORK", "id":22943, "ctx":"listener","msg":"Connection accepted","attr":{"remote":"172.20.0.3:57078","isLoadBalanced":false,"uuiid":{"$uuiid":{"$uuiid":"12a64bd1-329b-4801-afbe-a059fc06ddf2"}}},"connectionId":3,"connectionCount":3}}
mongo_db | {"t":{"$date":"2026-04-19T07:18:32.133+00:00"},"s":"I", "c":"NETWORK", "id":51800, "ctx":"conn3","msg":"client metadata","attr":{"remote":"172.20.0.3:57078","client":"conn3","negotiatedCompressors":[],"doc":{"driver":{"name":"nodejs|Mongoose","version":"7.1.1|9.4.1"},"platform":"Node.js v18.20.8, LE","os":{"name":"linux","architecture":"x64","version":"6.6.87.2-microsoft-standard-WSL2"},"type":"Linux"},"env":{"container":{"runtime":"docker"}}}}}
mongo_db | {"t":{"$date":"2026-04-19T07:18:32.133+00:00"},"s":"I", "c":"ACCESS", "id":10483900,"ctx":"conn3","msg":"Connection not authenticating","attr":{"client":"172.20.0.3:57078","doc":{"driver":{"name":"nodejs|Mongoose","version":"7.1.1|9.4.1"},"platform":"Node.js v18.20.8, LE","os":{"name":"linux","architecture":"x64","version":"6.6.87.2-microsoft-standard-WSL2"},"type":"Linux"},"env":{"container":{"runtime":"docker"}}}}}
mongo_db | {"t":{"$date":"2026-04-19T07:19:22.317+00:00"},"s":"I", "c":"WTCHKPT", "id":22430, "ctx":"Checkpoint", "msg":"WiredTiger message","attr":{"message":{"ts_sec":1776583162,"ts_usec":317338,"thread":"1:0x7b1e473bd6c0","session_name":"WT_SESSION.checkpoint","category":"WT_VERB_CHECKPOINT_PROGRESS","log_id":1000000,"category_id":7,"verbose_level":"INFO","verbose_level_id":0,"msg":"saving checkpoint snapshot min: 34, snapshot max: 34 snapshot count: 0, oldest timestamp: (0, 0), meta checkpoint timestamp: (0, 0) base write gen: 1"}}}}
mongo_db | {"t":{"$date":"2026-04-19T07:20:23.354+00:00"},"s":"I", "c":"WTCHKPT", "id":22430, "ctx":"Checkpoint", "msg":"WiredTiger message","attr":{"message":{"ts_sec":1776583223,"ts_usec":354533,"thread":"1:0x7b1e473bd6c0","session_name":"WT_SESSION.checkpoint","category":"WT_VERB_CHECKPOINT_PROGRESS","log_id":1000000,"category_id":7,"verbose_level":"INFO","verbose_level_id":0,"msg":"saving checkpoint snapshot min: 35, snapshot max: 35 snapshot count: 0, oldest timestamp: (0, 0), meta checkpoint timestamp: (0, 0) base write gen: 1"}}}}
```

Output



Pratical-2- Wordpress

```
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\harik>D:

D:\>mkdir wordpress-docker

D:\>cd wordpress-docker

D:\wordpress-docker>touch docker-compose.yml
'touch' is not recognized as an internal or external command,
operable program or batch file.

D:\wordpress-docker>notepad docker-compose.yml

D:\wordpress-docker>docker-compose up -d
time="2026-04-19T12:55:11+05:30" level=warning msg="D:\\wordpress-docker\\docker-compose.yml: the attribute 'version' is obsolete, it
will be ignored, please remove it to avoid potential confusion"
[+] up 2/2
✔Container wordpress_db Recreated 1.6s
✔Container wordpress_app Recreated 1.9s

D:\wordpress-docker>docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
4b3e97cf56b4   wordpress:latest                    "docker-entrypoint.s..." 14 seconds ago Up 12 seconds 0.0.0.0:8080->80/tcp, [::]:8080
ec8aa137f4b8   mysql:5.7                          "docker-entrypoint.s..." 16 seconds ago Up 12 seconds 3306/tcp, 33060/tcp
fdf089da4cf7   microservices-demo-backend         "docker-entrypoint.s..." 7 minutes ago  Up 7 minutes  0.0.0.0:3000->3000/tcp, [::]:3000
10eabd1fd604   mongo                              "docker-entrypoint.s..." 7 minutes ago  Up 7 minutes  0.0.0.0:27017->27017/tcp, [::]:27017
27017->27017/tcp mongo_db
f795b1f6d44d   kindest/node:v1.35.0              "/usr/local/bin/entr..." 29 hours ago   Up 15 minutes 127.0.0.1:60140->6443/tcp
zero-cloud-cluster-control-plane

D:\wordpress-docker>docker-compose down
time="2026-04-19T12:57:43+05:30" level=warning msg="D:\\wordpress-docker\\docker-compose.yml: the attribute 'version' is obsolete, it
will be ignored, please remove it to avoid potential confusion"
[+] down 3/3
✔Container wordpress_app Removed 1.4s
✔Container wordpress_db Removed 1.5s
✔Network wordpress-docker_default Removed 0.4s

D:\wordpress-docker>docker-compose up -d
time="2026-04-19T12:58:01+05:30" level=warning msg="D:\\wordpress-docker\\docker-compose.yml: the attribute 'version' is obsolete, it
will be ignored, please remove it to avoid potential confusion"
[+] up 3/3
✔Network wordpress-docker_default Created 0.0s
✔Container wordpress_db Created 0.1s
✔Container wordpress_app Created 0.2s

D:\wordpress-docker>docker logs wordpress_db
2026-04-19 07:28:02+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 5.7.44-1.el7 started.
2026-04-19 07:28:02+00:00 [Note] [Entrypoint]: Switching to dedicated user 'mysql'
2026-04-19 07:28:02+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 5.7.44-1.el7 started.
'/var/lib/mysql/mysql.sock' -> '/var/run/mysqld/mysqld.sock'
2026-04-19T07:28:02.860105Z 0 [Warning] TIMESTAMP with implicit DEFAULT value is deprecated. Please use --explicit_defaults_for_timestamp server option (see documentation for more details).
2026-04-19T07:28:02.861467Z 0 [Note] mysqld (mysqld 5.7.44) starting as process 1 ...
2026-04-19T07:28:02.864878Z 0 [Note] InnoDB: PUNCH HOLE support available
2026-04-19T07:28:02.864914Z 0 [Note] InnoDB: Mutexes and rw_locks use GCC atomic builtins
2026-04-19T07:28:02.864917Z 0 [Note] InnoDB: Uses event mutexes
2026-04-19T07:28:02.864919Z 0 [Note] InnoDB: GCC builtin __atomic_thread_fence() is used for memory barrier
2026-04-19T07:28:02.864920Z 0 [Note] InnoDB: Compressed tables use zlib 1.2.13
2026-04-19T07:28:02.864922Z 0 [Note] InnoDB: Using Linux native AIO
```

```
time="2026-04-19T13:00:08+05:30" level=warning msg="D:\\wordpress-docker\\docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] up 4/4
✓Network wordpress-docker_default Created 0.0s
✓Volume wordpress-docker_db_data Created 0.0s
✓Container wordpress_db Created 0.1s
✓Container wordpress_app Created 0.3s
```

```
D:\wordpress-docker>docker network ls
NETWORK ID          NAME                DRIVER              SCOPE
20c14cf309ce        bridge              bridge               local
ac92920dd3c3        host                host                 local
0c5f0cf1f2a1        kind                bridge               local
b3ede7ed56e4        microservices-demo_default bridge              local
0a96f61d2116        none                null                 local
1f4ddc4ebe3c        wordpress-docker_default bridge              local
```

```
D:\wordpress-docker>docker network inspect wordpress-docker_default
[
  {
    "Name": "wordpress-docker_default",
    "Id": "1f4ddc4ebe3c93219a53641dff982945a331b47e1dc3b2ee53c4a922d6f53cdc",
    "Created": "2026-04-19T07:30:09.130849873Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": null,
      "Config": [
        {
          "Subnet": "172.18.0.0/16",
          "Gateway": "172.18.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "ConfigOnly": false,
    "Options": {
      "com.docker.network.enable_ipv4": "true",
      "com.docker.network.enable_ipv6": "false"
    },
    "Labels": {
      "com.docker.compose.config-hash": "442430a8675e9a503bdc7758e43f01058ff51905b85c541066969534ead2951",
      "com.docker.compose.network": "default",
      "com.docker.compose.project": "wordpress-docker",
      "com.docker.compose.version": "5.0.2"
    },
    "Containers": {
      "a0b75c0105ff14bee0db730bbe4de535d55da724d2e6146982c24e92219df82a": {
        "Name": "wordpress_db",
        "EndpointID": "897beb43794c095e77499dc2975992fd272a48f5995b45a4163736df7fe3a1d7",
        "MacAddress": "86:04:42:0f:69:20",
        "IPv4Address": "172.18.0.2/16",
        "IPv6Address": ""
      },
      "ebc3c867b68f05ca5d2ba77025f0ef79b9c41bcf61835c393b6b6eb5fb80b40": {
        "Name": "wordpress_app",
        "EndpointID": "772fdd566f9eea1021eb34b4d1fc70ec5042f36e213a3f80d354b8e2b8ec86e4",
        "MacAddress": "82:72:ea:65:40:1f",
        "IPv4Address": "172.18.0.3/16",
        "IPv6Address": ""
      }
    },
    "Status": {
      "IPAM": {
        "Subnets": [
          {
            "172.18.0.0/16": {
              "IPsInUse": 5,
              "DynamicIPsAvailable": 65531
            }
          }
        ]
      }
    }
  }
]
```

```
},
"ConfigOnly": false,
"Options": {
  "com.docker.network.enable_ipv4": "true",
  "com.docker.network.enable_ipv6": "false"
},
"Labels": {
  "com.docker.compose.config-hash": "442430a8675e9a503bdc7758e43f01058ff51905b85c541066969534ead2951",
  "com.docker.compose.network": "default",
  "com.docker.compose.project": "wordpress-docker",
  "com.docker.compose.version": "5.0.2"
},
"Containers": {
  "a0b75c0105ff14bee0db730bbe4de535d55da724d2e6146982c24e92219df82a": {
    "Name": "wordpress_db",
    "EndpointID": "897beb43794c095e77499dc2975992fd272a48f5995b45a4163736df7fe3a1d7",
    "MacAddress": "86:04:42:0f:69:20",
    "IPv4Address": "172.18.0.2/16",
    "IPv6Address": ""
  },
  "ebc3c867b68f05ca5d2ba77025f0ef79b9c41bcf61835c393b6b6eb5fb80b40": {
    "Name": "wordpress_app",
    "EndpointID": "772fdd566f9eea1021eb34b4d1fc70ec5042f36e213a3f80d354b8e2b8ec86e4",
    "MacAddress": "82:72:ea:65:40:1f",
    "IPv4Address": "172.18.0.3/16",
    "IPv6Address": ""
  }
},
"Status": {
  "IPAM": {
    "Subnets": [
      {
        "172.18.0.0/16": {
          "IPsInUse": 5,
          "DynamicIPsAvailable": 65531
        }
      }
    ]
  }
}
}
```

```

2026-04-19T07:30:20.378373Z 0 [Note] InnoDB: GCC builtin __atomic_thread_fence() is used for memory barrier
2026-04-19T07:30:20.378375Z 0 [Note] InnoDB: Compressed tables use zlib 1.2.13
2026-04-19T07:30:20.378377Z 0 [Note] InnoDB: Using Linux native AIO
2026-04-19T07:30:20.378733Z 0 [Note] InnoDB: Number of pools: 1
2026-04-19T07:30:20.378892Z 0 [Note] InnoDB: Using CPU crc32 instructions
2026-04-19T07:30:20.381516Z 0 [Note] InnoDB: Initializing buffer pool, total size = 128M, instances = 1, chunk size = 128M
2026-04-19T07:30:20.398410Z 0 [Note] InnoDB: Completed initialization of buffer pool
2026-04-19T07:30:20.403802Z 0 [Note] InnoDB: If the mysqld execution user is authorized, page cleaner thread priority can be changed.
See the man page of setpriority().
2026-04-19T07:30:20.415726Z 0 [Note] InnoDB: Highest supported file format is Barracuda.
2026-04-19T07:30:20.427342Z 0 [Note] InnoDB: Creating shared tablespace for temporary tables
2026-04-19T07:30:20.427678Z 0 [Note] InnoDB: Setting file './ibtmp1' size to 12 MB. Physically writing the file full; Please wait ...
2026-04-19T07:30:20.445420Z 0 [Note] InnoDB: File './ibtmp1' size is now 12 MB.
2026-04-19T07:30:20.446299Z 0 [Note] InnoDB: 96 redo rollback segment(s) found. 96 redo rollback segment(s) are active.
2026-04-19T07:30:20.446334Z 0 [Note] InnoDB: 32 non-redo rollback segment(s) are active.
2026-04-19T07:30:20.447202Z 0 [Note] InnoDB: 5.7.44 started; log sequence number 12219419
2026-04-19T07:30:20.447509Z 0 [Note] InnoDB: Loading buffer pool(s) from /var/lib/mysql/ib_buffer_pool
2026-04-19T07:30:20.449622Z 0 [Note] Plugin 'FEDERATED' is disabled.
2026-04-19T07:30:20.452221Z 0 [Note] InnoDB: Buffer pool(s) load completed at 260419 7:30:20
2026-04-19T07:30:20.457061Z 0 [Note] Found ca.pem, server-cert.pem and server-key.pem in data directory. Trying to enable SSL support
using them.
2026-04-19T07:30:20.457087Z 0 [Note] Skipping generation of SSL certificates as certificate files are present in data directory.
2026-04-19T07:30:20.457090Z 0 [Warning] A deprecated TLS version TLSv1 is enabled. Please use TLSv1.2 or higher.
2026-04-19T07:30:20.457091Z 0 [Warning] A deprecated TLS version TLSv1.1 is enabled. Please use TLSv1.2 or higher.
2026-04-19T07:30:20.457548Z 0 [Warning] CA certificate ca.pem is self signed.
2026-04-19T07:30:20.457606Z 0 [Note] Skipping generation of RSA key pair as key files are present in data directory.
2026-04-19T07:30:20.457860Z 0 [Note] Server hostname (bind-address): '*'; port: 3306
2026-04-19T07:30:20.457927Z 0 [Note] IPv6 is available.
2026-04-19T07:30:20.457935Z 0 [Note] - '::' resolves to '::';
2026-04-19T07:30:20.457947Z 0 [Note] Server socket created on IP: '::'.
2026-04-19T07:30:20.461496Z 0 [Warning] Insecure configuration for --pid-file: Location '/var/run/mysqld' in the path is accessible t
o all OS users. Consider choosing a different directory.
2026-04-19T07:30:20.468139Z 0 [Note] Event Scheduler: Loaded 0 events
2026-04-19T07:30:20.469144Z 0 [Note] mysqld: ready for connections.
Version: '5.7.44' socket: '/var/run/mysqld/mysqld.sock' port: 3306 MySQL Community Server (GPL)

D:\wordpress-docker>docker restart wordpress_app
wordpress_app

D:\wordpress-docker>

```

Output-

Practical no. 4 : Docker Comp x Microservices with Docker Co x localhost:3000 x WordPress - Installation x

localhost:8080/wp-admin/install.php?step=1

Welcome

Welcome to the famous five-minute WordPress installation process! Just fill in the information below and you'll be on your way to using the most extendable and powerful personal publishing platform in the world.

Information needed

Please provide the following information. Do not worry, you can always change these settings later.

Site Title

Username
Usernames can have only alphanumeric characters, spaces, underscores, hyphens, periods, and the @ symbol.

Password [Hide](#)
Strong

Important: You will need this password to log in. Please store it in a secure location.

Your Email
Double-check your email address before continuing.

Search engine visibility ☒ Discourage search engines from indexing this site
It is up to search engines to honor this request.

[Install WordPress](#)

32° Search ENG IN 13:06 19-04-2026

Practical no. 4 : Docker Comp x Microservices with Docker Co x localhost:3000 x Posts - Hari.tech - WordPress x

localhost:8080/wp-admin/edit.php

Hari.tech 0 + New Howdy, harikrishna_777

WordPress 6.9.4 is available! [Please update now.](#)

Posts [Add Post](#)

All (1) | Published (1)

Bulk actions [Apply](#) All dates All Categories [Filter](#) 1 item

☐	Title	Author	Categories	Tags		Date
☐	Hello world!	harikrishna_777	Uncategorized	—	1	Published 2026/04/19 at 7:37 am
☐	Title	Author	Categories	Tags		Date

Bulk actions [Apply](#) 1 item

Thank you for creating with [WordPress](#). [Get Version 6.9.4](#)

Practical-4 Deploy Node.js + MongoDB using Docker Compose

```
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\harik>D:

D:\>mkdir node-mongo-compose

D:\>

D:\>cd node-mongo-compose

D:\node-mongo-compose>notepad server.js

D:\node-mongo-compose>notepad package.json

D:\node-mongo-compose>notepad Dockerfile

D:\node-mongo-compose>ren Dockerfile.txt Dockerfile

D:\node-mongo-compose>notepad docker-compose.yml

D:\node-mongo-compose>docker compose up --build
time="2026-04-19T13:27:58+05:30" level=warning msg="D:\\node-mongo-compose\\docker-compose.yml: the attribute 'version' is obsolete,
it will be ignored, please remove it to avoid potential confusion"
#1 [internal] load local bake definitions
#1 reading from stdin 507B done
#1 DONE 0.0s

#2 [internal] load build definition from Dockerfile
#2 transferring dockerfile: 161B done
#2 DONE 0.1s

#3 [internal] load metadata for docker.io/library/node:18
#3 ...

#4 [auth] library/node:pull token for registry-1.docker.io
#4 DONE 0.0s
```

```

#9 [4/5] RUN npm install
#9 CACHED

#10 [5/5] COPY .
#10 CACHED

#11 exporting to image
#11 exporting layers done
#11 exporting manifest sha256:6bb634ef29fa3b7edff8da24a7f57fd878fb680f8408784ca2d1ca41ad8612da done
#11 exporting config sha256:334ceb02daeb9582c6323e422386e699724f271e9b5470041e140fcd28855019 done
#11 exporting attestation manifest sha256:7066971a16e34730fda3b26e54be559b27c6f46d3a2ba6368efe476832ef42e 0.0s done
#11 exporting manifest list sha256:c0430a58bd177ee75408546ab38b9c87cfa2c59840f9446706cf2eaf31689de1 0.0s done
#11 naming to docker.io/library/node-mongo-compose-app:latest done
#11 unpacking to docker.io/library/node-mongo-compose-app:latest
#11 unpacking to docker.io/library/node-mongo-compose-app:latest 0.0s done
#11 DONE 0.2s

#12 resolving provenance for metadata file
#12 DONE 0.0s
[+] up 2/2
✔Image node-mongo-compose-app Built 5.1s
✔Container node_app Created 0.1s
Attaching to mongo_db, node_app
mongo_db | {"t":{"$date":"2026-04-19T08:01:12.916+00:00"},"s":"I", "c":"CONTROL", "id":23285, "ctx":"main","msg":"Automatically
disabling TLS 1.0, to force-enable TLS 1.0 specify --sslDisabledProtocols 'none'"}
mongo_db | {"t":{"$date":"2026-04-19T08:01:12.923+00:00"},"s":"I", "c":"NETWORK", "id":4915701, "ctx":"main","msg":"Initialized wi
re specification", "attr":{"spec":{"incomingExternalClient":{"minWireVersion":0,"maxWireVersion":17},"incomingInternalClient":{"minWir
eVersion":0,"maxWireVersion":17},"outgoing":{"minWireVersion":6,"maxWireVersion":17},"isInternalClient":true}}}
mongo_db | {"t":{"$date":"2026-04-19T08:01:12.925+00:00"},"s":"I", "c":"NETWORK", "id":4648601, "ctx":"main","msg":"Implicit TCP F
astOpen unavailable. If TCP FastOpen is required, set tcpFastOpenServer, tcpFastOpenClient, and tcpFastOpenQueueSize."}
mongo_db | {"t":{"$date":"2026-04-19T08:01:12.931+00:00"},"s":"I", "c":"REPL", "id":5123008, "ctx":"main","msg":"Successfully r
egistered PrimaryOnlyService", "attr":{"service":"TenantMigrationDonorService","namespace":"config.tenantMigrationDonors"}}
mongo_db | {"t":{"$date":"2026-04-19T08:01:12.931+00:00"},"s":"I", "c":"REPL", "id":5123008, "ctx":"main","msg":"Successfully r
egistered PrimaryOnlyService", "attr":{"service":"TenantMigrationRecipientService","namespace":"config.tenantMigrationRecipients"}}
mongo_db | {"t":{"$date":"2026-04-19T08:01:12.931+00:00"},"s":"I", "c":"REPL", "id":5123008, "ctx":"main","msg":"Successfully r
egistered PrimaryOnlyService", "attr":{"service":"ShardSplitDonorService","namespace":"config.tenantSplitDonors"}}
mongo_db | {"t":{"$date":"2026-04-19T08:01:12.931+00:00"},"s":"I", "c":"CONTROL", "id":5945603, "ctx":"main","msg":"Multi threadin
g initialized"}
mongo_db | {"t":{"$date":"2026-04-19T08:01:12.932+00:00"},"s":"I", "c":"CONTROL", "id":4615611, "ctx":"initandlisten","msg":"Mongo
DB starting", "attr":{"pid":1,"port":27017,"dbPath":"/data/db","architecture":"64-bit","host":"0710d40bbb41"}}

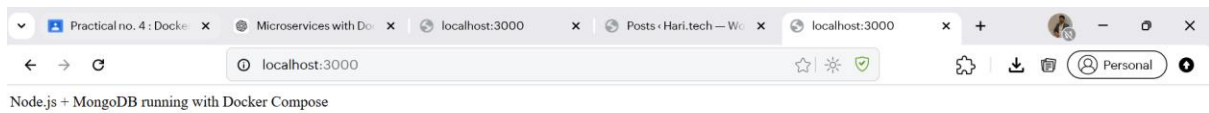
--91521151082749432323", "commitTimestamp":null}}
mongo_db | {"t":{"$date":"2026-04-19T08:01:13.729+00:00"},"s":"I", "c":"NETWORK", "id":22943, "ctx":"listener","msg":"Connection
accepted", "attr":{"remote":"172.21.0.3:57088","uuid":"25f86626-dd16-4f10-901d-66f93c2533b3","connectionId":1,"connectionCount":1}}

mongo_db | {"t":{"$date":"2026-04-19T08:01:13.737+00:00"},"s":"I",
"c":"NETWORK", "id":51800, "ctx":"conn1","msg":"client metadata", "attr":{"remote":"172.21.0.3:57088","client":{"conn1","negotiated
Compressors":[],"doc":{"driver":{"name":"nodejs|Mongoose","version":"5.9.2|7.8.9"},"platform":"Node.js v18.20.8, LE","os":{"name":"li
nux","architecture":"x64","version":"6.6.87.2-microsoft-standard-WSL2"},"type":"Linux"}}}}
node_app | MongoDB Connected

View in Docker Desktop View Config Enable Watch Detmongo_db | {"t":{"$date":"2026-04-19T08:01:24.247+00:00"},"s":"I",
"c":"NETWORK", "id":22943, "ctx":"listener","msg":"Connection accepted", "attr":{"remote":"172.21.0.3:34702","uuid":"c073141a-f673
-48b2-9fd0-2d224d26ee8f","connectionId":2,"connectionCount":2}}
mongo_db | {"t":{"$date":"2026-04-19T08:01:24.248+00:00"},"s":"I", "c":"NETWORK", "id":51800, "ctx":"conn2","msg":"client metada
ta", "attr":{"remote":"172.21.0.3:34702","client":{"conn2","negotiatedCompressors":[],"doc":{"driver":{"name":"nodejs|Mongoose","versio
n":"5.9.2|7.8.9"},"platform":"Node.js v18.20.8, LE","os":{"name":"linux","architecture":"x64","version":"6.6.87.2-microsoft-standard-
WSL2"},"type":"Linux"}}}}
mongo_db | {"t":{"$date":"2026-04-19T08:02:13.230+00:00"},"s":"I", "c":"WTCHKPT", "id":22430, "ctx":"Checkpoint", "msg":"WiredT
iger message", "attr":{"message":{"ts_sec":1776585733,"ts_usec":230264,"thread":"1:0x7be2969ee640","session_name":"WT_SESSION.checkpoi
nt","category":"WT_VERB_CHECKPOINT_PROGRESS","category_id":6,"verbose_level":"DEBUG","verbose_level_id":1,"msg":"saving checkpoint sn
apshot min: 33, snapshot max: 33 snapshot count: 0, oldest timestamp: (0, 0), meta checkpoint timestamp: (0, 0) base write gen: 1"}}
}
mongo_db | {"t":{"$date":"2026-04-19T08:03:13.269+00:00"},"s":"I", "c":"WTCHKPT", "id":22430, "ctx":"Checkpoint", "msg":"WiredT
iger message", "attr":{"message":{"ts_sec":1776585793,"ts_usec":269229,"thread":"1:0x7be2969ee640","session_name":"WT_SESSION.checkpoi
nt","category":"WT_VERB_CHECKPOINT_PROGRESS","category_id":6,"verbose_level":"DEBUG","verbose_level_id":1,"msg":"saving checkpoint sn
apshot min: 36, snapshot max: 36 snapshot count: 0, oldest timestamp: (0, 0), meta checkpoint timestamp: (0, 0) base write gen: 1"}}
}
mongo_db | {"t":{"$date":"2026-04-19T08:04:13.298+00:00"},"s":"I", "c":"WTCHKPT", "id":22430, "ctx":"Checkpoint", "msg":"WiredT
iger message", "attr":{"message":{"ts_sec":1776585853,"ts_usec":298198,"thread":"1:0x7be2969ee640","session_name":"WT_SESSION.checkpoi
nt","category":"WT_VERB_CHECKPOINT_PROGRESS","category_id":6,"verbose_level":"DEBUG","verbose_level_id":1,"msg":"saving checkpoint sn
apshot min: 38, snapshot max: 38 snapshot count: 0, oldest timestamp: (0, 0), meta checkpoint timestamp: (0, 0) base write gen: 1"}}
}

```

Output-



Pratical-5 Appache compose

```
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\harik>D:

D:\>mkdir apache-compose1

D:\>cd apache-compose1

D:\apache-compose1>mkdir website

D:\apache-compose1>cd website

D:\apache-compose1\website>notepad index.html

D:\apache-compose1\website>

D:\apache-compose1\website>cd ..

D:\apache-compose1>notepad docker-compose.yml

D:\apache-compose1>docker compose up -d
[+] up 2/2
  ✓Network apache-compose1_default Created 0.0s
  ✓Container apache_server Created 0.2s

D:\apache-compose1>docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
4762aa026715   httpd                                "httpd-foreground"      7 seconds ago Up 6 seconds  0.0.0.0:8081->80/tcp, [::]:8081->80/tcp
59b6e21c9b64   node-mongo-compose-app             "docker-entrypoint.s..." 10 minutes ago Up 10 minutes  0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
0710d40bbb41   node_app                            "docker-entrypoint.s..." 12 minutes ago Up 10 minutes  0.0.0.0:27017->27017/tcp, [::]:27017->27017/tcp
7017->27017/tcp mongo_db                             "docker-entrypoint.s..." 41 minutes ago Up 38 minutes  0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
ebc3c867b68f   wordpress:latest                   "docker-entrypoint.s..." 41 minutes ago Up 41 minutes  3306/tcp, 33060/tcp
a0b75c0105ff   mysql:5.7                          "docker-entrypoint.s..." 30 hours ago   Up About an hour  127.0.0.1:60140->6443/tcp
f795b1f6d44d   kindest/node:v1.35.0              "/usr/local/bin/entr..." 30 hours ago   Up About an hour  127.0.0.1:60140->6443/tcp
```

Practical no. 4: x Microservices wi x localhost:3000 x Posts (Hari.tech) x localhost:3000 x localhost:8081 x + Personal

localhost:8081

Hello from Apache Docker Compose

Pratical-6 Nginx + Redis

```
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\harik>D:

D:\>mkdir nginx-redis-compose

D:\>cd nginx-redis-compose

D:\nginx-redis-compose>notepad docker-compose.yml

D:\nginx-redis-compose>docker compose up -d
[+] up 13/13
✔Image redis Pulled 17.1s
✔Network nginx-redis-compose_appnetwork Created 0.1s
✔Container nginx_server Created 0.3s
✔Container nginx-redis-compose-redis-1 Created 0.3s

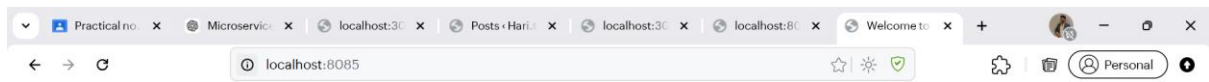
D:\nginx-redis-compose>docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS
0466b2a3816f nginx "/docker-entrypoint..." 14 seconds ago Up 13 seconds 0.0.0.0:8085->80/tcp, [::]:8085->80/tcp
84251c18a229 redis "docker-entrypoint.s..." 14 seconds ago Up 13 seconds 6379/tcp
4762aa026715 httpd "httpd-foreground" 6 minutes ago Up 6 minutes 0.0.0.0:8081->80/tcp, [::]:8081->80/tcp
59b6e21c9b64 node-mongo-compose-app "docker-entrypoint.s..." 17 minutes ago Up 17 minutes 0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
0710d40bbb41 mongo:6 "docker-entrypoint.s..." 18 minutes ago Up 17 minutes 0.0.0.0:27017->27017/tcp, [::]:27017->27017/tcp
ebc3c867b68f wordpress:latest "docker-entrypoint.s..." 48 minutes ago Up 45 minutes 0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
a0b75c0105ff mysql:5.7 "docker-entrypoint.s..." 48 minutes ago Up 48 minutes 3306/tcp, 33060/tcp
f795b1f6d44d kindest/node:v1.35.0 "/usr/local/bin/entr..." 30 hours ago Up About an hour 127.0.0.1:60140->6443/tcp
zero-cloud-cluster-control-plane

D:\nginx-redis-compose>
```

zero-cloud-cluster-control-plane

```
D:\nginx-redis-compose>docker network ls
NETWORK ID NAME DRIVER SCOPE
10325ef47c88 apache-compose1_default bridge local
20c14cf309ce bridge bridge local
ac92920dd3c3 host host local
0c5f0cf1f2a1 kind bridge local
b3ede7ed56e4 microservices-demo_default bridge local
e05820423533 nginx-redis-compose_appnetwork bridge local
3f0ca625b3ce node-mongo-compose_default bridge local
0a96f61d2116 none null local
1f4ddc4ebe3c wordpress-docker_default bridge local
```

Output



Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.